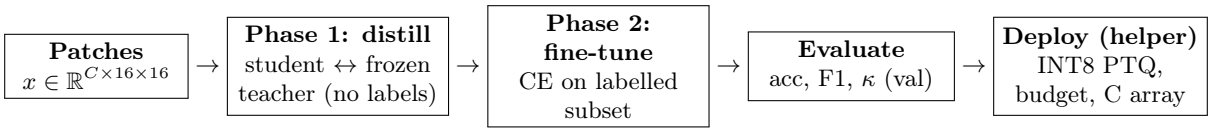
tinygfm: Research Overview

Tiny Multi-Task Network for On-Board AI Inference in Earth Observation, distilled from Geospatial Foundation Models

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1 Scope and status

`tinygfm` is a PyTorch scaffold that tests one question: *at equal model size (≤100 KB of INT8 weights), when does a student CNN distilled from a frozen Geospatial Foundation Model (GFM) beat the same student trained conventionally?* Factors varied: student size, label fraction, spectral bands, baseline type, single vs. multi-task head.



Pipeline: distil, fine-tune, evaluate, deploy.

2 Dataset

Training has two phases. In Phase 1 the student imitates the teacher on unlabelled Sentinel-2 patches, so it can use a large and diverse pool; in Phase 2 the heads are fine-tuned on the labelled patches of CloudSEN12, Sen1Floods11 and THRawS.

2.1 SSL4EO-S12 (unlabelled Sentinel-2 pool for distillation)

It has 251,079 locations, mostly around large cities, each seen in four seasons, with images of 264 × 264 px (2.64 km at 10 m) and Sentinel-2 L1C (13 bands). That gives about 1 M images, or 2.6×10^8 patches of 16×16[†]; a random subset of a few hundred thousand images across seasons and locations is enough.

Points to check.

- *Few clouds.* Only scenes with under 10% cloud cover are kept, so the unlabelled CloudSEN12 images are added to the pool for the cloud task.
- *Disjoint areas.* Pool images that overlap the validation or test areas of the three datasets are removed.
- *Teacher overlap.* Several foundation models were pre-trained on SSL4EO-S12, so teacher and student may see the same images; this is declared.

2.2 CloudSEN12 (cloud and cloud shadow, control task)

Content. Global Sentinel-2 scenes (2018–2020). Each region of interest (ROI) is 5,090 × 5,090 m at 10 m, so one image patch (IP) is ≈ 509 × 509 px, and each ROI has several IPs with different cloud cover. High-quality labels: 2,000 ROIs = 10,000 IPs; no label: ≈29,250 IPs. Classes: clear, thick cloud, thin cloud, cloud shadow. Data: Sentinel-2 L1C (13 bands) and L2A, plus Sentinel-1 and DEM.

Labels and split. 14 trained experts, with human agreement of 95.7% (thin cloud is the weak spot, about 78%); a U-Net with MobileNetV2 reaches 0.92 balanced accuracy, versus 0.84 for Fmask. Spatial hexagon split: 1,827 training and 173 test ROIs; all IPs of an ROI stay in the same split.

2.3 Sen1Floods11 (surface water)

Content. 11 flood events, 4,831 chips of 512 × 512 px at 10 m. Sentinel-2 L1C (13 bands, UInt16, reflectance ×10,000) and Sentinel-1 (VV, VH); labels −1 no data, 0 not water, 1 water.

Labels and split. 446 *hand-labelled* chips, 4,385 *weakly labelled* chips (automatic rules, no quality control) and 814 JRC chips. Hand-labelled split: 252 train, 89 validation, 90 test, plus the Bolivia event held out as a geographic test (15 chips[†]).

2.4 THRawS (thermal hotspots in raw Sentinel-2)

Content. Raw (Level-0) Sentinel-2 granules built for on-board processing: only B8A, B11, B12 at 20 m; 88 events (58 volcanic, 20 wildfires, 10 hard negatives).

Labels. Bounding boxes, not masks, produced by a simplified Massimetti et al. algorithm and checked by two experts. Patches of 128 × 128 px with 25% overlap: 1,090 hotspot vs. 33,335 event-free (3.17% positive). No official split and no benchmark results.

2.5 Patch counts and memory at 16×16

Cutting each image into non-overlapping 16×16 patches gives the counts below. One patch takes $4\,C \cdot 256$ bytes in float32 (13,312 B with 13 bands, 3,072 B with 3 bands), and half of that as UInt16.

Subset	images	patches†	float32†	UInt16†
Sen1Floods11, hand-labelled	446	456,704	6.1 GB	3.0 GB
Sen1Floods11, all chips (weak labels)	4,831	4,946,944	65.9 GB	32.9 GB
CloudSEN12, high-quality	10,000	9,610,000	127.9 GB	64.0 GB
CloudSEN12, no label	29,250	28,109,250	374 GB	187 GB
THRawS (C=3)	34,425	≤2,203,200	6.8 GB	3.4 GB

Patch counts and storage per dataset (derived estimate).

The totals are large, so a subsample of about 50–100 k patches per dataset and split is enough for the experiments, and the unlabelled CloudSEN12 images are not needed in full. THRawS counts are an upper bound, since its patches overlap by 25%. Labelled fractions are always drawn by whole chip, image or event, never by single patch, because neighbouring patches look almost identical and would leak labels between training and test.

2.6 Why these three datasets, and what each one tests

The thesis trains one tiny network on three on-board decisions that share the same input type but differ in physics and class balance.

	CloudSEN12	**Sen1Floods11**	**THRawS**
Patch label	cloud / not cloud	water / not water	hotspot / none
Sensor data	Sentinel-2, 10 m	Sentinel-2, 10 m	raw Sentinel-2, 20 m
Bands	13	13	3 (B8A, B11, B12)
Ground truth	pixel masks	pixel masks	bounding boxes
Positive patches	moderate	low–moderate	very rare (3.17%)
Role	control task	medium difficulty	hardest task

The three datasets compared.

2.7 From masks and boxes to patch labels

The student sees a 16×16 patch P and outputs one class. CloudSEN12 and Sen1Floods11 give a label for every pixel; THRawS gives bounding boxes, which we turn into a pixel mask ($m(p) = 1$ inside a box, 0 outside). The patch label is then computed from the mask of its 256 pixels.

Cloud and water. Let $m(p) = 1$ if pixel p belongs to the target class (cloud = thick or thin; water) and 0 otherwise, ignoring invalid pixels (e.g. label -1). With V the set of valid pixels in P, we compute the valid fraction v and the covered fraction ρ:

$$v = \frac{|V|}{256}, \qquad \rho = \frac{1}{|V|} \sum_{p \in V} m(p). \tag{1}$$

The patch is discarded if $v < v_{\min}$; otherwise

$$y = \begin{cases} 1 & \rho \geq \tau \\ 0 & \rho < \tau. \end{cases} \tag{2}$$

The thresholds τ and $v_{\min}$ are fixed once and reported; a higher τ gives fewer but cleaner positives, and we report results for $\tau \in \{0.05, 0.25, 0.5\}$.

Hotspots. A hotspot can cover only a few pixels, so we count them instead of using a fraction: the patch is positive if

$$\sum_{p \in P} m(p) \geq n_{\min}, \tag{3}$$

and negative otherwise. A box is larger than the burning area, so these labels are coarser than the pixel masks; $n_{\min}$ is chosen on the validation set. At 20 m a patch covers 320 m (160 m at 10 m for the other datasets), so THRawS is reported separately. Without overlap, an $H \times W$ image gives $N = \lfloor H/16 \rfloor \lfloor W/16 \rfloor$ patches.

2.8 Splits and leakage: evaluation protocol

Neighbouring 16×16 patches are almost the same image, so a split made patch by patch lets the test set "see" the training set and inflates every result. The protocol below is proposed for the thesis; only the official splits come from the dataset sources.

Split by group. All patches of the same group (ROI, chip or event) always go to the same split, so training, validation and test share no group. Official splits are used where they exist.

Dataset	Group	Split
CloudSEN12	ROI	Official: 1,827 train / 173 test ROIs. Validation taken from train, by ROI.
Sen1Floods11	Event	Official: 252 train / 89 val / 90 test chips.
THRawS	Event	None official. Own split by event, with volcanoes, wildfires and non-events in every split.

Split rules for each dataset.

Labels, distillation and statistics. Smaller label fractions (e.g. 1, 5, 10, 100%) are nested subsets of whole groups, identical for the distilled model and for every baseline at a given seed. The unlabelled pool for Phase 1 never includes validation or test images. Band statistics are computed on training data only, the decision threshold is set on validation, and the test set is evaluated once.

Metrics. Accuracy is misleading when positives are rare (at 3.17% positives, predicting always "no hotspot" scores 96.8%), so we report the true positive rate at a fixed false positive rate (set on validation), plus F1 and precision–recall AUC. For each seed and label fraction we compare distilled and baseline models on the same test set, $\Delta = M^{\text{distill}} - M^{\text{baseline}}$, and estimate its 95% interval by resampling test *groups*, not patches, 1,000 times. A result counts as an improvement only if the interval excludes 0.

Leakage checks. (a) no group appears in two splits and no duplicate patch crosses splits; (b) with shuffled training labels the test performance must fall to chance; (c) one experiment is repeated with a random patch-level split, and the gap to the group-level result shows how optimistic a naive split would be. The foundation-model teachers may have seen the test areas during pre-training; this cannot be removed, so it is declared and applies equally to distilled models and baselines.

2.9 Data preparation and teacher targets

Using CloudSEN12, Sen1Floods11 and THRawS requires three choices, described below.

1. Data preparation. Each image is cut into 16×16 patches. For every patch we keep its group (ROI, chip or event), its position, the valid fraction v and the covered fraction ρ (or the hotspot pixel count), so that the labelling thresholds τ and $n_{\min}$ can be changed without cutting the images again. Splits and label fractions follow the group-level protocol of Section 2.8.

2. Bands and normalisation. Bands are selected by name, not by position. THRawS has only B8A, B11 and B12, whereas the teacher expects more bands (Prithvi-EO-2.0 uses six: blue, green, red, narrow NIR, SWIR1, SWIR2), so the missing ones are either filled with zeros, and this is reported, or the teacher is given a reduced view. Sentinel-2 L1C values are reflectance scaled by 10,000, while THRawS raw values are not reflectance, so each band is normalised with statistics computed on training data only.

3 Teacher models

A geospatial foundation model (GFM) is a large network pre-trained without labels on millions of satellite images. In this thesis the GFM is only the teacher: it runs on the ground during training, produces one feature vector per 16×16 patch, and the tiny student learns to reproduce these features. The GFM is never sent to the satellite. Two candidates are compared: Prithvi-EO-2.0 (NASA and IBM) and Copernicus-FM (Wang et al., 2025).

3.1 Candidate models

Prithvi-EO-2.0 is a ViT trained as a masked autoencoder on 4.2 M samples of the Harmonized Landsat–Sentinel-2 (HLS) archive at 30 m, 2014–2023. Each sample is a 224×224 crop with up to 4 dates. It always takes the same six bands: blue, green, red, NIR, SWIR1, SWIR2.

Copernicus-FM is a ViT-Base trained on Copernicus-Pretrain (Sentinel-1, -2, -3, -5P and DEM). Its first layer is built from the wavelength of each band, so it accepts any set of bands; Sentinel-2 inputs are 264×264 px with 16×16 patches at 10 m.

	Prithvi-EO-2.0	Copernicus-FM
Size	300 M or 600 M	ViT-Base (≈86 M)
Bands	6, fixed	any
Resolution	30 m	10 m
Patch size	16 (300 M), 14 (600 M)	16
Input size	224 × 224	264 × 264
Licence	open weights	CC BY 4.0

The two candidate teachers.

3.2 Compatibility with the pipeline

The thesis uses Sentinel-2 at 10 m, 16 × 16 patches, 13 bands for cloud and water, and only 3 bands (B8A, B11, B12) for hotspots.

Requirement	Prithvi-EO-2.0	Copernicus-FM
One token per 16 × 16 patch	only 300 M	yes
All 13 bands	no (6 of 13)	yes
Only 3 bands (THRawS)	no (3 of 6 missing)	yes
10 m resolution	no (30 m)	yes
Same size as SSL4EO-S12	no (224 crop)	yes (264)
Teacher cost	high	low

Fit of each teacher with the pipeline.

4 Student networks

5 Feature distillation

6 Multi-task fine-tuning

7 Quantisation and on-board constraints

8 Experimental design and evaluation

9 Results and analysis

10 Discussion, limitations and future work

Sources

Chapter 2: Dataset.
CloudSEN12: https://cloudsen12.github.io/; Aybar et al., *CloudSEN12, a global dataset for semantic understanding of cloud and cloud shadow in Sentinel-2*, Scientific Data 2022 (https://pmc.ncbi.nlm.nih.gov/articles/PMC9789947).
Sen1Floods11: https://github.com/cloudtostreet/Sen1Floods11; Bonafilia et al., CVPR Workshops 2020, pp. 210–211 (https://openaccess.thecvf.com/content_CVPRW_2020/papers/w11/Bonafilia_Sen1Floods11_A_Georeferenced_Dataset_to_Train_and_Test_Deep_Learning_CVPRW_2020_paper.pdf); split counts from arXiv:2302.08180.
THRawS: Zenodo https://zenodo.org/records/7908728 (DOI 10.5281/zenodo.7908728; code: ESA-PhiLab/PyRawS); Del Prete et al., *Unlocking the use of raw multispectral Earth observation imagery for onboard AI*, arXiv:2305.11891.
SSL4EO-S12 (unlabelled pool): Wang et al., *SSL4EO-S12: a large-scale multi-modal, multi-temporal dataset for self-supervised learning in Earth observation*, arXiv:2211.07044.

Chapter 3: Teacher models.
Prithvi-EO-2.0: Szwarcman et al., *Prithvi-EO-2.0: a versatile multi-temporal foundation model for Earth observation applications*, arXiv:2412.02732 (weights: Hugging Face, `ibm-nasa-geospatial/Prithvi-EO-2.0`).
Copernicus-FM: Wang et al., *Towards a unified Copernicus foundation model for Earth vision*, arXiv:2503.11849.